

ABSTRACT

Legal documents such as rental agreements, employment contracts, loan papers and notices are written in complex language that most people find hard to understand.

This often leads to signed agreements with hidden risks and unfair terms. Hiring a lawyer for every document is costly and slow, especially for students, tenants, freelancers and small business owners.

This project presents the *Legal Document Analyser and Adviser*, a web-based application developed in Java using the Spring Boot framework.

Users can upload legal documents in PDF, DOCX, TXT or scanned image format. The system extracts the text using Apache PDFBox, Apache POI and Tesseract (OCR), and processes it with Natural Language Processing techniques and a Large Language Model (LLM).

It then classifies the document type and generates a plain-language summary. It identifies key clauses and entities such as parties, dates, payment terms, termination conditions and governing law. It also flags unfair, ambiguous or missing clauses with Low, Medium or High severity and explains difficult legal terms in simple words.

A question-and-answer adviser built on Retrieval-Augmented Generation (RAG) lets users ask follow-up questions and receive answers grounded in their own document, so the system does not invent clauses.

It also suggests next steps and negotiation points, and users can download the full analysis as a PDF report.

The application uses Spring Security with JWT for authentication and role-based access, Spring Data JPA with Hibernate for persistence, and PostgreSQL or MySQL for storage. It includes a user history module and an administrator dashboard.

The system is for informational assistance only and does not replace a qualified lawyer. By making legal documents understandable and highlighting risks early, it helps ordinary users make informed decisions.

Keywords: Legal Document Analysis, Java, Spring Boot, NLP, Large Language Model, RAG, OCR, Risk Detection